

CREATE Module: Perturbation Theory Lecture 6: Boundary Layers in Fluid Mechanics



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Constant density Navier-Stokes Equations

Momentum

$$\frac{D\vec{u}}{Dt} = -\frac{1}{\rho_0} \nabla p + \nu \nabla^2 \vec{u}$$

Dimensional
Form

Mass

$$\vec{\nabla} \cdot \vec{u} = 0$$

By nondimensionalizing the Navier Stokes equations we create something that looks a bit like a perturbation theory problem. In this set of slides we explore to what extent this is the case.

$$\frac{D\vec{u}}{Dt} = -\nabla p + \frac{1}{Re} \nabla^2 \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

Dimensionless
Form

Non-dimensionalizing the Navier Stokes Equations

$$\rho_0 [u_t + uu_x + wu_z] = -p_x + \mu \nabla^2 u$$

$$\vec{u} = U \tilde{u} \ , \ (x, z) = L(\tilde{x}, \tilde{z}) \ , \ p = \rho_0 U^2 \tilde{p} \ , \ t = \frac{L}{U} \tilde{t}$$

$$\rho_0 \frac{U^2}{L} [\tilde{u}_{\tilde{t}} + \tilde{u}\tilde{u}_{\tilde{x}} + \tilde{w}\tilde{u}_{\tilde{z}}] = \rho_0 \frac{U^2}{L} (-\tilde{p}_{\tilde{x}}) + \frac{\mu U}{L^2} \tilde{\nabla}^2 \tilde{u}$$

divide through, let $Re = \frac{UL}{\nu}$ and drop tildes.

$$u_t + uu_x + wu_z = -p_x + \frac{1}{Re} \nabla^2 u$$

$Re \rightarrow \infty$ means we get Euler equations

The Reynolds Number

- ❖ The Reynolds number is the most famous dimensionless quantity in fluid mechanics.
- ❖ It represents the ratio of inertia (or nonlinear) terms in the governing to the viscous (or dissipative) terms in the governing equations.
- ❖ The size of the Reynolds number is set by a physical parameter, viscosity, which is measured in the lab, and two flow dependent parameters, the characteristic length and the characteristic velocity.
- ❖ This means we can have moderate viscosity but low Reynolds number if we are considering slow flow on tiny scales like the zooplankton shown below.



Low and High Reynolds Number



- ❖ For a tiny copepod in a lake a typical length $L \approx 10^{-3}m$
- ❖ A typical velocity might be $U \approx 10^{-3}ms^{-1}$
- ❖ The viscosity of water is $\nu \approx 10^{-6}m^2s^{-1}$
- ❖ So $Re \approx \frac{10^{-3} \times 10^{-3}}{10^{-6}} = 1$



- ❖ For air flow over the math building $L \approx 3 \times 10^1m$
- ❖ A typical velocity might be $U \approx 10^1ms^{-1}$
- ❖ The viscosity of air is very similar to water, $\nu \approx 10^{-6}m^2s^{-1}$
- ❖ So $Re \approx \frac{10 \times 30}{10^{-6}} = 3 \times 10^8$

Low Reynolds Number

$$Re \left(\frac{D\vec{u}}{Dt} + \nabla p \right) = \nabla^2 \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

$$\tilde{p} = \frac{p}{\rho_0 U^2}$$

- ❖ In the $Re \rightarrow 0$ limit things reduce to several Laplace's equations.
- ❖ With no slip BCs would get zero velocity everywhere!
- ❖ This is because there can't be a balance between two different mechanisms (say viscosity balancing pressure)

- ❖ So we must rescale the pressure:

$$\tilde{p} = \frac{p}{\mu U/L}$$

Notice the scaling
now depends on
viscosity

Low Reynolds Number

$$Re \frac{D\vec{u}}{Dt} = -\nabla p + \nabla^2 \vec{u}$$

- ❖ In the $Re \rightarrow 0$ limit things reduce to several Poisson type equations so that time appears as a parameter.
- ❖ With no slip BCs and conservation of mass we would get

$$\begin{aligned}\nabla p &= \nabla^2 \vec{u} \\ \nabla \cdot \vec{u} &= 0\end{aligned}$$

This is an example of dominant balance that you saw in the singular problem for polynomials

- ❖ At the boundary either get no slip conditions $\vec{u} = 0$ or an imposed force $\tau_{ij}n_j = -pn_i + 2\mu e_{ij}n_j = F_i^{(applied)}$
- ❖ But either way the equations are linear! A strange world for most people in the quantitative climate sciences.

Summary

- ❖ The Reynolds number allows to unambiguously compare the nonlinear (or inertia) terms to the viscous terms.
- ❖ High Reynolds number is associated with fast flows, large scales and not very viscous fluids.
- ❖ In the $Re \rightarrow \infty$ limit the Navier Stokes equations reduce to the Euler equations.
- ❖ This is a singular limit because the highest spatial derivatives drop out.
- ❖ The low Reynolds number limit requires a rescaling of pressure in order to get a set of equations that has non zero solutions.
- ❖ The equations in this limit are linear, and time only appears as a parameter, with some amazing consequences.
- ❖ https://www.youtube.com/watch?v=po8_KlTKP5o

Overview

- ❖ We have covered the basic scaling of fluid mechanics and the process of non-dimensionalization.
- ❖ Next we introduce a simple, fundamental idea having to do with the theory of boundary layers: the concept of different length scales in different directions.
- ❖ Historically, this is the work of Prandtl (who shockingly never won a Nobel prize for it), but in the modern context a more obvious context is the fluid mechanics of the atmosphere and ocean; for both of these horizontal scales can extend to a thousand kilometers or more, but the vertical extent of the ocean is typically on the order of a few kilometers, and the layer of the atmosphere which we inhabit is about ten kilometers thick.
- ❖ Since large scale motions are also influenced by the Earth's rotation we will stick to Prandtl's work in these slides, but return to the concept later in the module when we discuss rotating flows.

Constant density Navier-Stokes Equations

Momentum

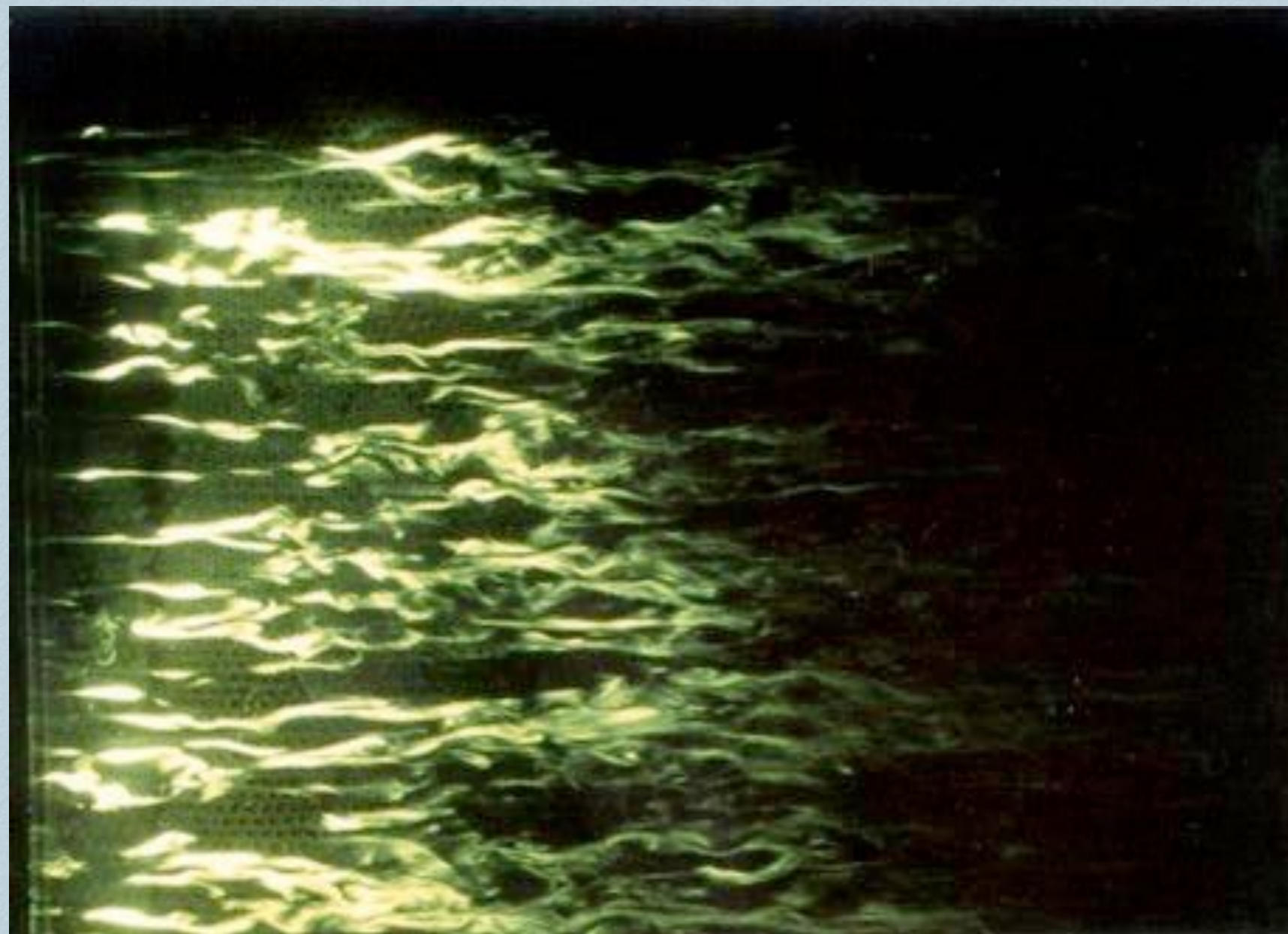
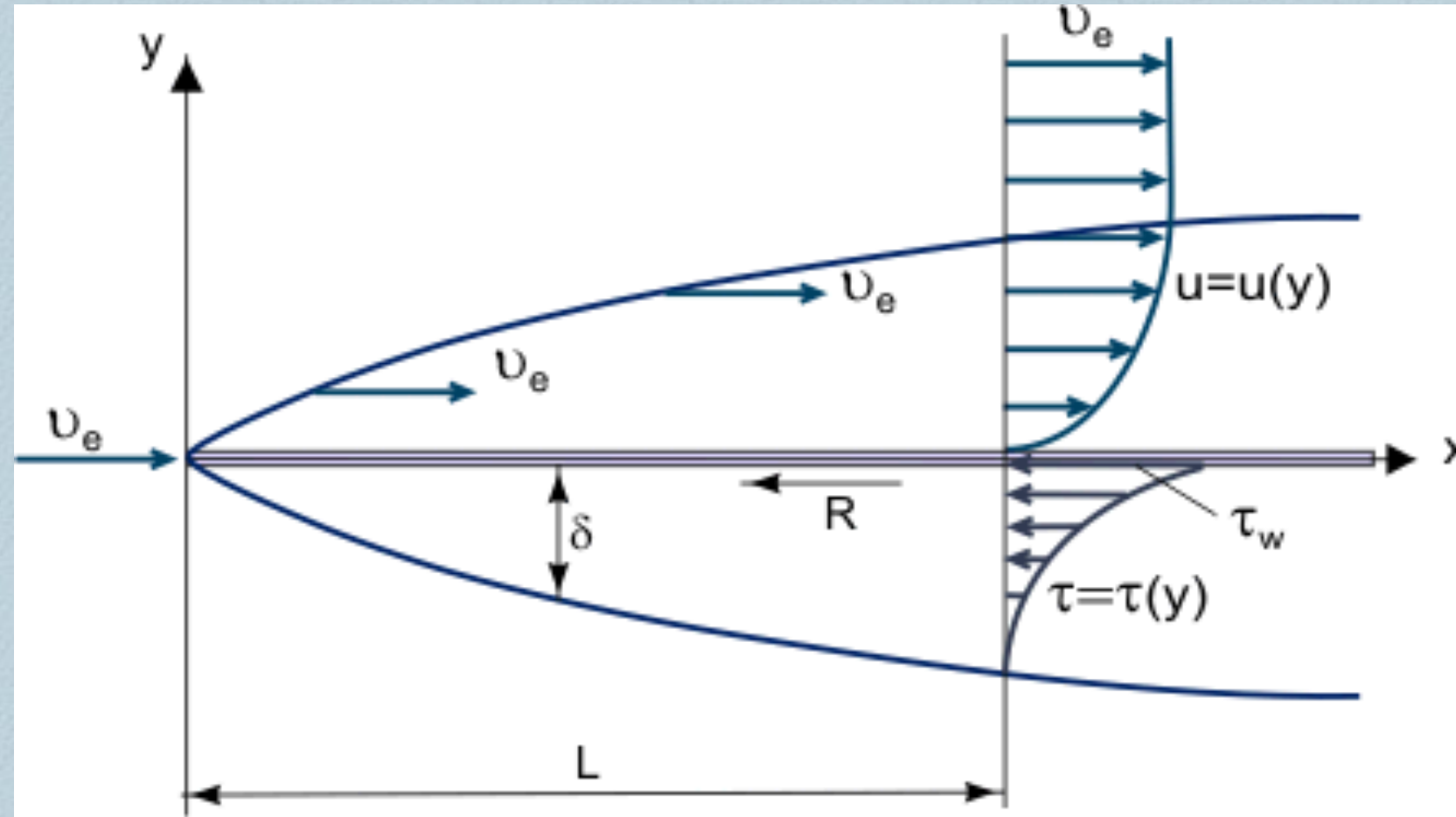
$$\frac{D\vec{u}}{Dt} = -\frac{1}{\rho_0} \nabla p + \nu \nabla^2 \vec{u}$$

Mass

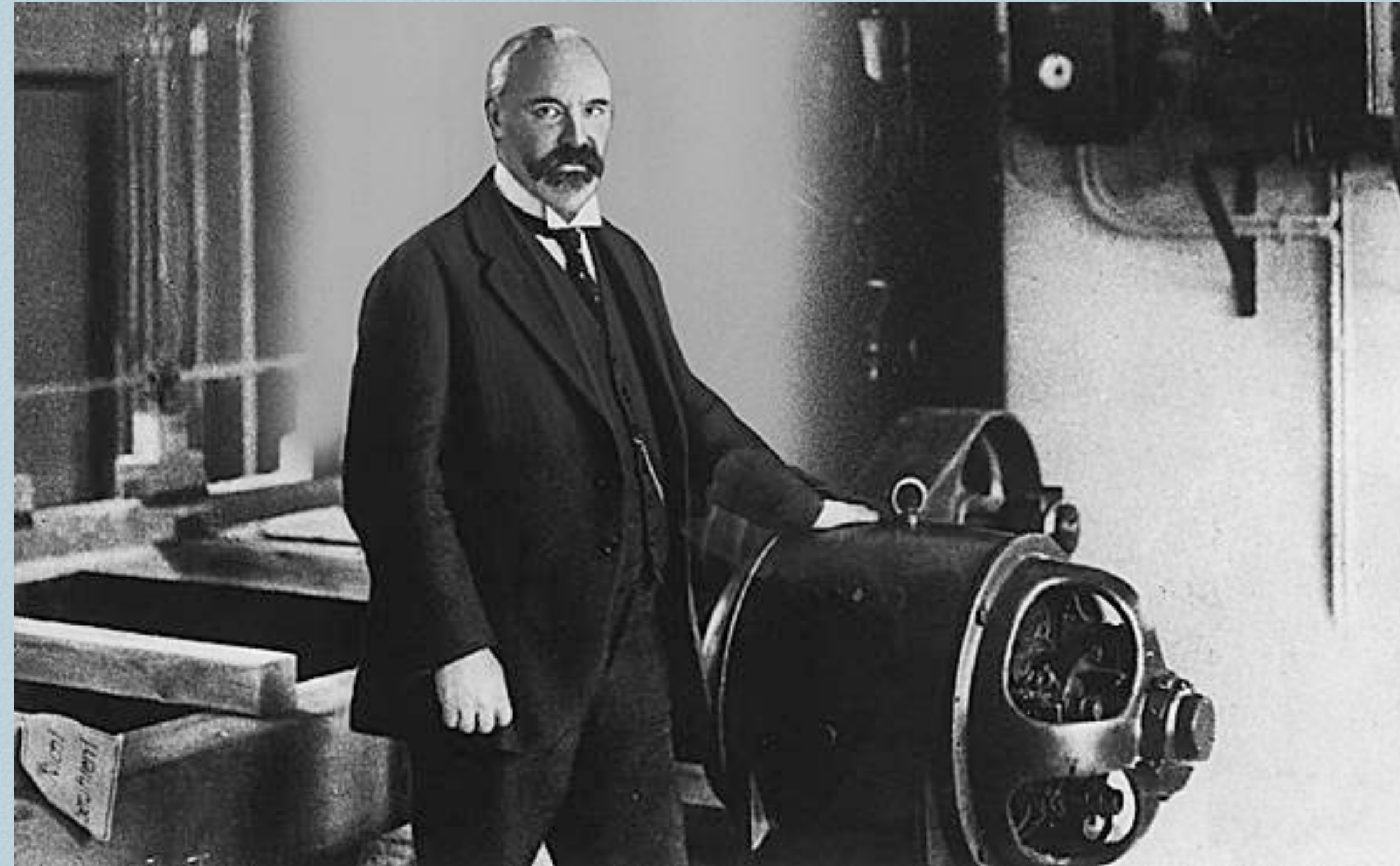
$$\vec{\nabla} \cdot \vec{u} = 0$$

For most common fluids like air and water the viscosity is very small.

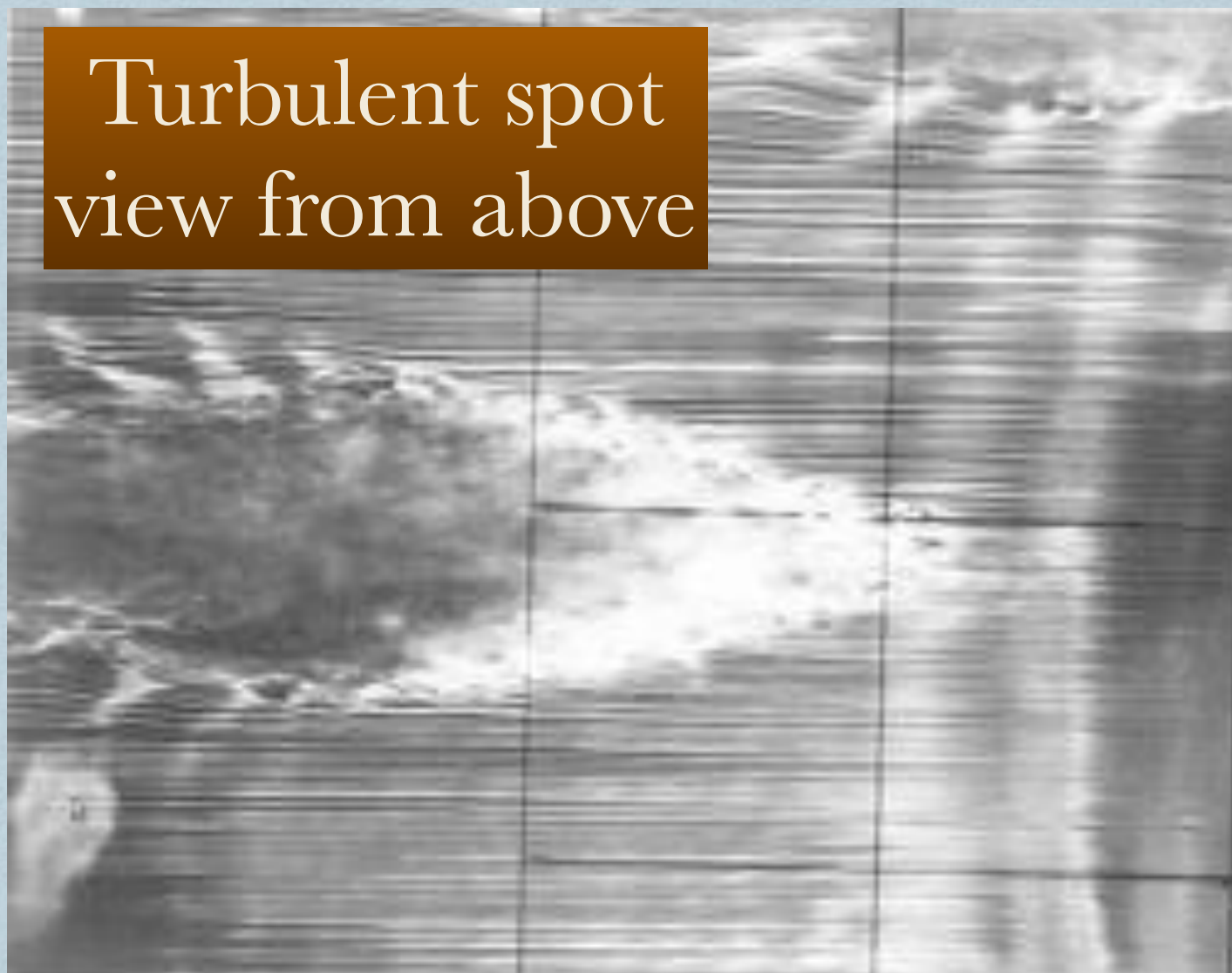
But dropping it entirely seems at odds with experiment since the no slip condition says velocities are zero at the solid boundaries.



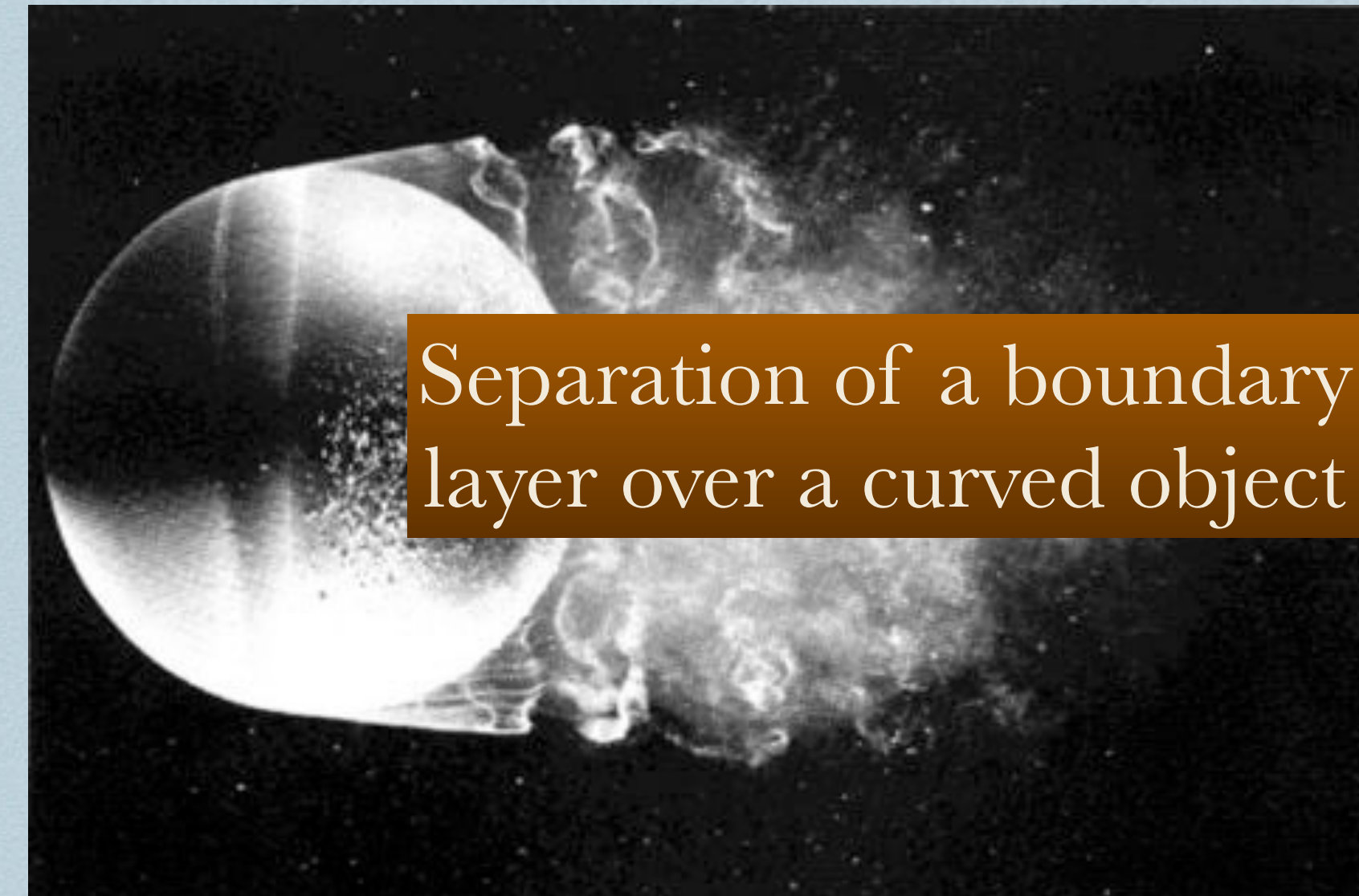
- ❖ Experiments suggest that viscosity matters only near solid boundaries when the flow is fast (the Reynolds number is high)
- ❖ This is an idea that goes back to Ludwig Prandtl (on next slide) at the turn of the 20th century.
- ❖ In practice the majority of boundary layers are turbulent like the picture on the left (top view of flow from left to right visualized by a bubble wire).
- ❖ Our goal in this set of slides, however, is to derive the simplest boundary layer set of equations for laminar flow (pictured above).



Prandtl
in the
lab



Turbulent spot
view from above



Separation of a boundary
layer over a curved object

- ❖ Boundary layers are thin regions near solid boundaries.
- ❖ They can be dealt with by using good quality numerical methods and high resolution, but historically people wanted a simplified set of equations.
- ❖ Let's consider the flat plate example.
- ❖ The key idea is to notice that “typical length” needs to be refined to “typical vertical length” and “typical horizontal length.
- ❖ First I review the traditional non-dimensionalization leading to the Reynolds number and then I proceed to boundary layer scaling.

$$\rho_0 [u_t + uu_x + wu_z] = -p_x + \mu \nabla^2 u$$

$$\vec{u} = U \tilde{u} \ , \ (x, z) = L(\tilde{x}, \tilde{z}) \ , \ p = \rho_0 U^2 \tilde{p} \ , \ t = \frac{L}{U} \tilde{t}$$

$$\rho_0 \frac{U^2}{L} [\tilde{u}_{\tilde{t}} + \tilde{u}\tilde{u}_{\tilde{x}} + \tilde{w}\tilde{u}_{\tilde{z}}] = \rho_0 \frac{U^2}{L} (-\tilde{p}_{\tilde{x}}) + \frac{\mu U}{L^2} \tilde{\nabla}^2 \tilde{u}$$

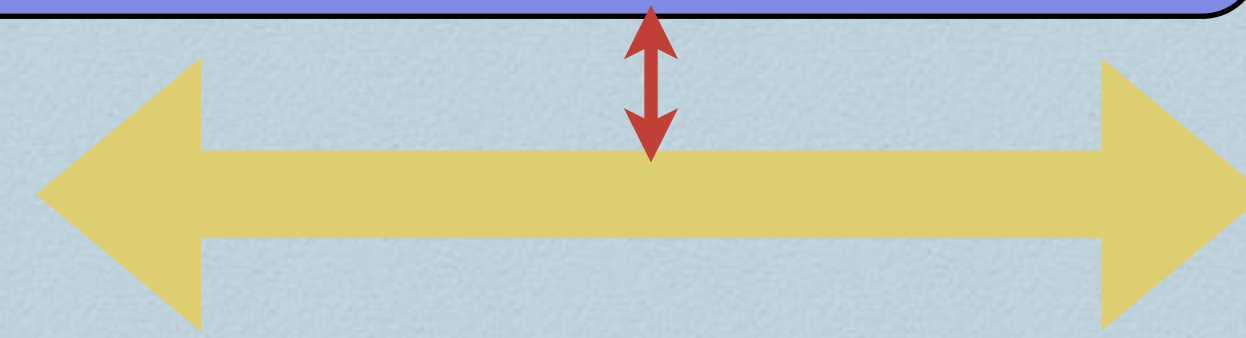
divide through, let $Re = \frac{UL}{\nu}$ and drop tildes.

$$u_t + uu_x + wu_z = -p_x + \frac{1}{Re} \nabla^2 u$$

$Re \rightarrow \infty$ means we get Euler equations

$$(u, w) = (U\tilde{u}, W\tilde{w}) \text{ , } (x, z) = (L\tilde{x}, H\tilde{z})$$

$$p = \rho_0 U^2 \tilde{p} \text{ , } t = \frac{L}{U} \tilde{t}$$



$$u_x + w_z = 0 \text{ implies } W = \frac{H}{L} U$$

$$\rho_0 [u_t + uu_x + wu_z] = -p_x + \mu [u_{xx} + u_{zz}]$$

$$\rho_0 \frac{U^2}{L} [\tilde{u}_{\tilde{t}} + \tilde{u}\tilde{u}_{\tilde{x}} + \tilde{w}\tilde{u}_{\tilde{z}}] = \rho_0 \frac{U^2}{L} (-\tilde{p}_{\tilde{x}}) + \frac{\mu U}{L^2} \tilde{u}_{\tilde{x}\tilde{x}} + \frac{\mu U}{H^2} \tilde{u}_{\tilde{z}\tilde{z}}$$

divide through, let $Re = \frac{UL}{\nu}$ and drop tildes.

$$u_t + uu_x + wu_z = -p_x + \frac{1}{Re} \left[u_{xx} + \frac{L^2}{H^2} u_{zz} \right]$$

But we can choose $\frac{L}{H} = \sqrt{Re}$ so that $H \ll L$

$$u_t + uu_x + wu_z = -p_x + u_{zz}$$

$$(u, w) = (U\tilde{u}, W\tilde{w}) , (x, z) = (L\tilde{x}, H\tilde{z})$$

$$p = \rho_0 U^2 \tilde{p} , t = \frac{L}{U} \tilde{t}$$

$$u_x + w_z = 0 \text{ implies } W = \frac{H}{L} U$$

$$\rho_0 [w_t + uw_x + ww_z] = -p_z + \mu [w_{xx} + w_{zz}]$$

$$\rho_0 \frac{UW}{L} [\tilde{w}_{\tilde{t}} + \tilde{u}\tilde{w}_{\tilde{x}} + \tilde{w}\tilde{u}_{\tilde{z}}] = \rho_0 \frac{U^2}{H} (-\tilde{p}_{\tilde{z}}) + \frac{\mu W}{L^2} \tilde{w}_{\tilde{x}\tilde{x}} + \frac{\mu W}{H^2} \tilde{w}_{\tilde{z}\tilde{z}}$$

$$\rho_0 U^2 \frac{H^2}{L^2} [\tilde{w}_{\tilde{t}} + \tilde{u}\tilde{w}_{\tilde{x}} + \tilde{w}\tilde{u}_{\tilde{z}}] = \rho_0 U^2 (-\tilde{p}_{\tilde{z}}) + \frac{\mu H^2 U}{L^3} \tilde{w}_{\tilde{x}\tilde{x}} + \frac{\mu U}{L} \tilde{w}_{\tilde{z}\tilde{z}}$$

divide through, let $Re = \frac{UL}{\nu}$ and drop tildes.

$$\frac{H^2}{L^2} [w_t + uw_x + ww_z] = -p_z + \frac{1}{Re} \left[\frac{H^2}{L^2} w_{xx} + w_{zz} \right]$$

But both $\frac{1}{Re}$ and $\frac{H}{L} \rightarrow 0$

So that we finally get $0 = -p_z$

$$u_t + uu_x + wu_z = -p_x + u_{zz}$$

$$0 = -p_z$$

$$u_x + w_z = 0$$

- ❖ The first equation states that horizontal momentum following a fluid particle is balanced by the horizontal pressure gradient and the **VERTICAL** diffusion of momentum ONLY.
- ❖ The second states that pressure is **CONSTANT** across the boundary layer.
- ❖ The third equation reaffirms incompressibility.
- ❖ The boundary layer equations are still nonlinear, but provide a great deal of physical clarity as to what is going on in the boundary layer.

- ❖ Next I want to return to a problem with an exact solution and reconsider it from the point of view of boundary layer theory.
- ❖ Consider the oscillating plate, under an infinite half-space of viscous fluid, with no pressure gradient in the horizontal.

$$u_t = \nu u_{zz}$$

$$u(z = 0, t) = \cos(\omega t)$$

$$u \rightarrow 0 \text{ as } z \rightarrow \infty$$

$$u(z, t) = \exp(-mz) \cos(\omega t + mz)$$

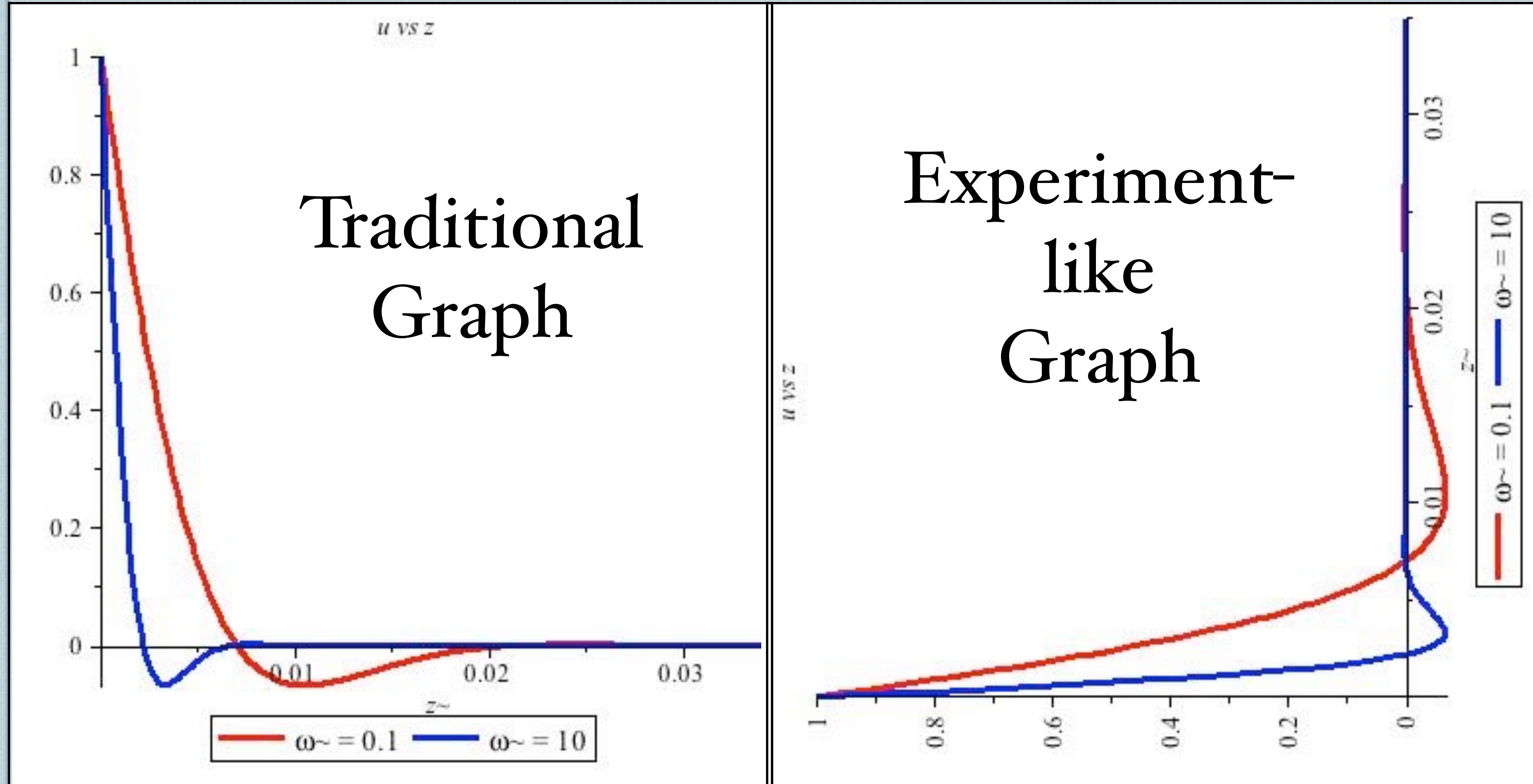
$$\text{where } m = \sqrt{\frac{\omega}{2\nu}}$$

- ❖ The solution gives oscillations in time AND in z , but they are damped in z . It is the exponential decay in z that sets the scale on which any appreciable motion takes place:

$$L_{BL} = \sqrt{\frac{2\nu}{\omega}}$$

- ❖ Note how BL thickness shrinks as the square root of frequency!

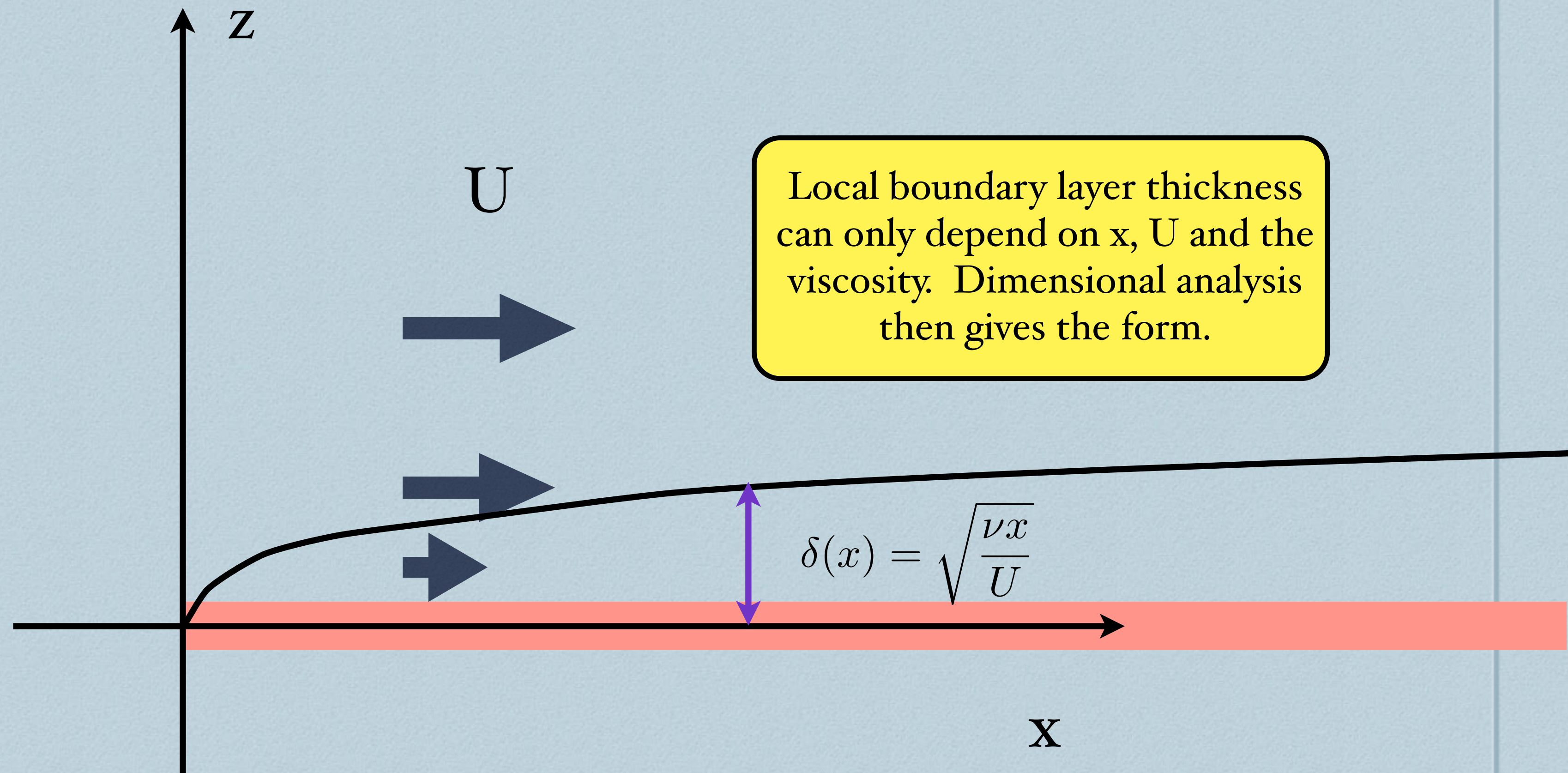
Sample solutions using water as the fluid.



The z axis is in meters

- ❖ There are other ways to measure the thickness of the boundary layer.
- ❖ All are motivated by various engineering applications.
- ❖ For example, you can measure the distance by which streamlines are displaced from their far upstream location. This is called displacement thickness.
- ❖ For a first treatment of boundary layers I don't want to dig too deep into these various details (though you can read about them in Kundu's book).
- ❖ I do want to return to the flat plate boundary layer and present the classical Blasius solution.
- ❖ This is an important problem because it illustrates the boundary layer concept in practice and because it does not have an analytical solution. In the notes I consider a boundary layer problem that does have a closed form solution.
- ❖ This latter point is one I make over and over again in my slides, stressing the need to take on the nonlinearity of fluid motion, instead of avoiding it as has been historical practice.

Blasius Solution for the Flat Plate Boundary Layer I



$$uu_x + wu_z = \nu u_{zzz} , \quad u_x + w_z = 0$$

$$u \rightarrow U \text{ as } z \rightarrow \infty$$

$$u(x=0, z) = Uu = w = 0 \text{ at } z = 0$$

Blasius Solution for the Flat Plate Boundary Layer II

$u = \psi_z$, $w = -\psi_x$ from incompressibility

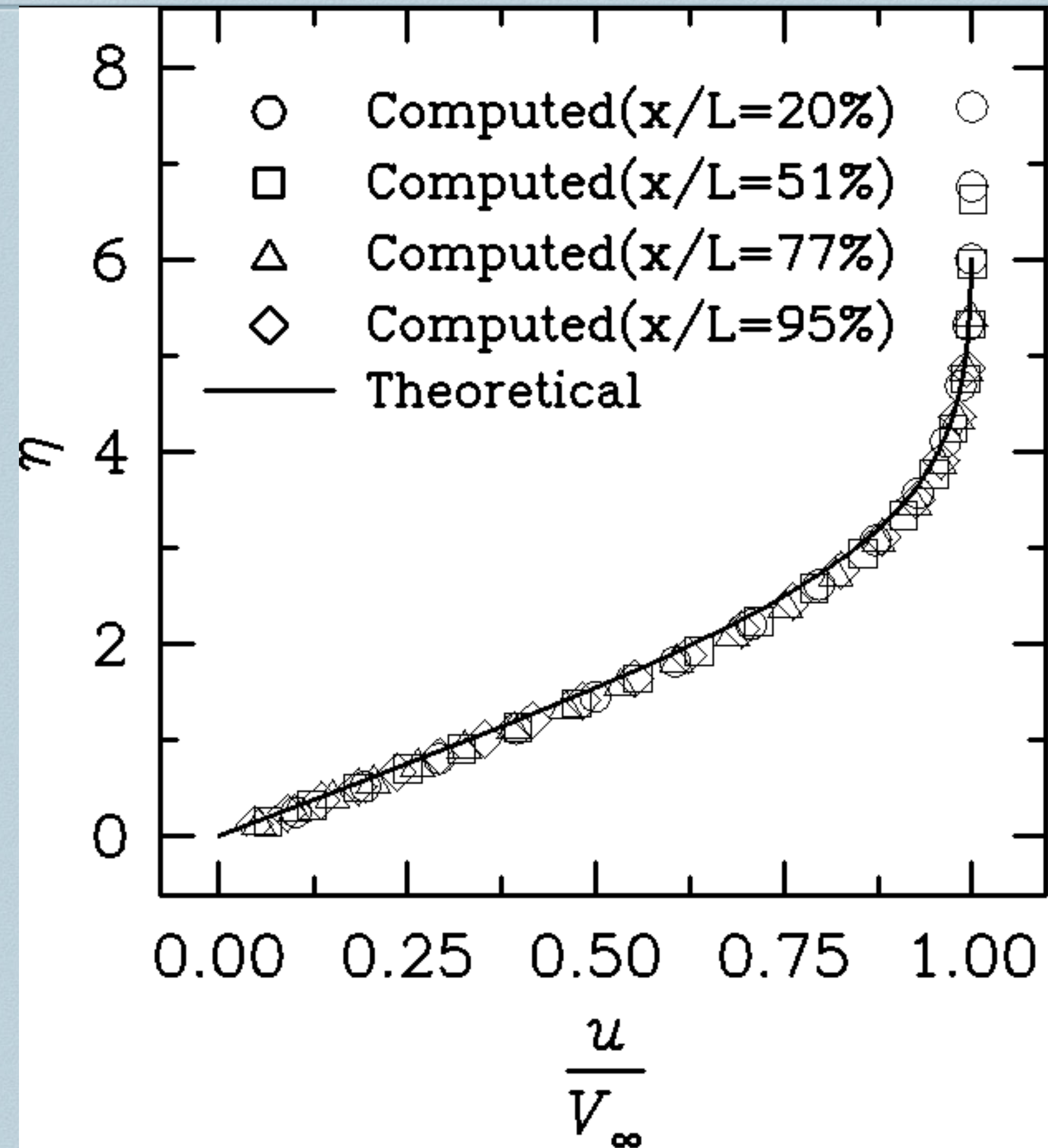
$u/U = g(\eta)$, $\eta = \frac{z}{\delta(x)}$ Similarity Solution

$g(\eta) = \frac{df}{d\eta}$ Trick to allow use of ψ

$\frac{1}{2} f f'' + f''' = 0$ After a lot of algebra

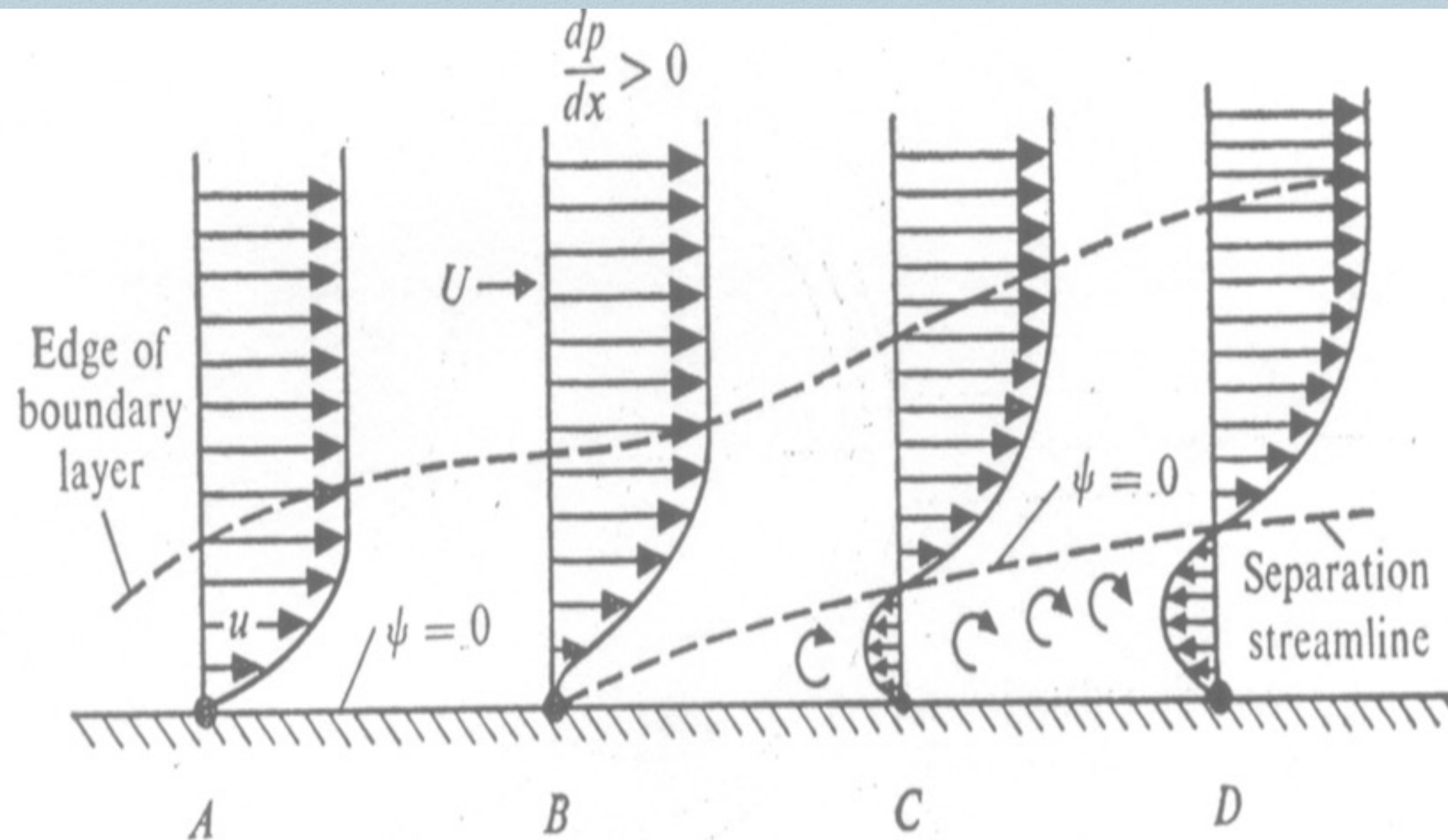
$f(0) = f'(0) = 0$, $f'(\eta) \rightarrow 1$ as $\eta \rightarrow \infty$

- ❖ The key point is that by choosing the spatial, or similarity, variable η correctly, the solution at ALL x coordinates collapses onto a single curve.
- ❖ This single profile is the solution of an ODE (a simpler problem).
- ❖ Unfortunately Blasius' ODE has no closed form solution.

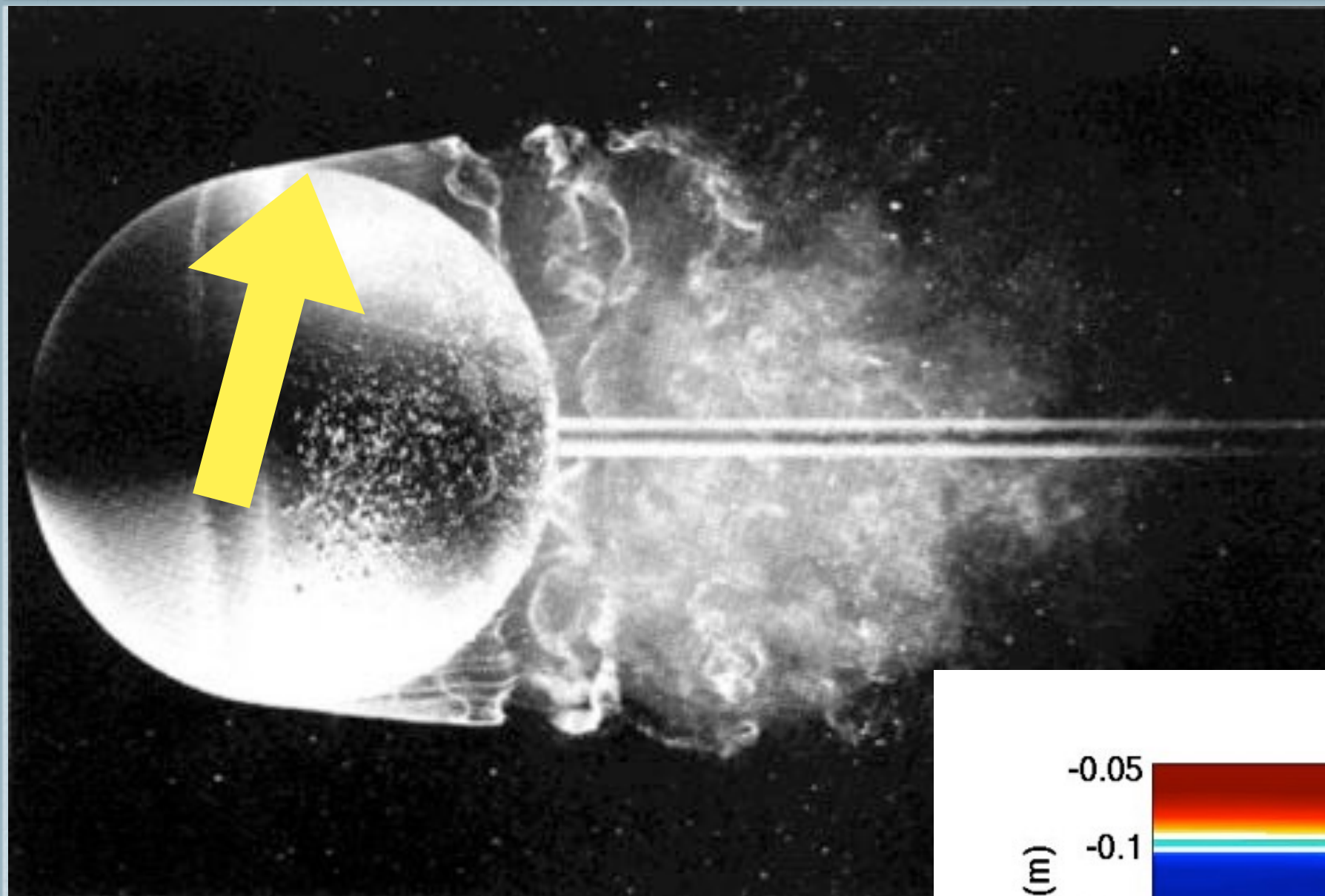


Good computing
exercise for lab book

- ❖ Let's compare and contrast the two boundary layer solutions we have looked at.
- ❖ For the oscillating plate the solution is time dependent, though in a very simple way, namely periodic and sinusoidal.
- ❖ For the oscillating plate the governing equations linearize (geometrically) and the solution is in closed form.
- ❖ For the oscillating plate the solution does not depend on x .
- ❖ For the Blasius boundary layer the equations are time independent and nonlinear, albeit simplified from the full Navier-Stokes equations.
- ❖ By defining a dimensionless variable the solution at all points collapses onto a single curve.
- ❖ However, this curve must be determined by approximate methods since the Blasius BVP for ODEs has no exact solution.

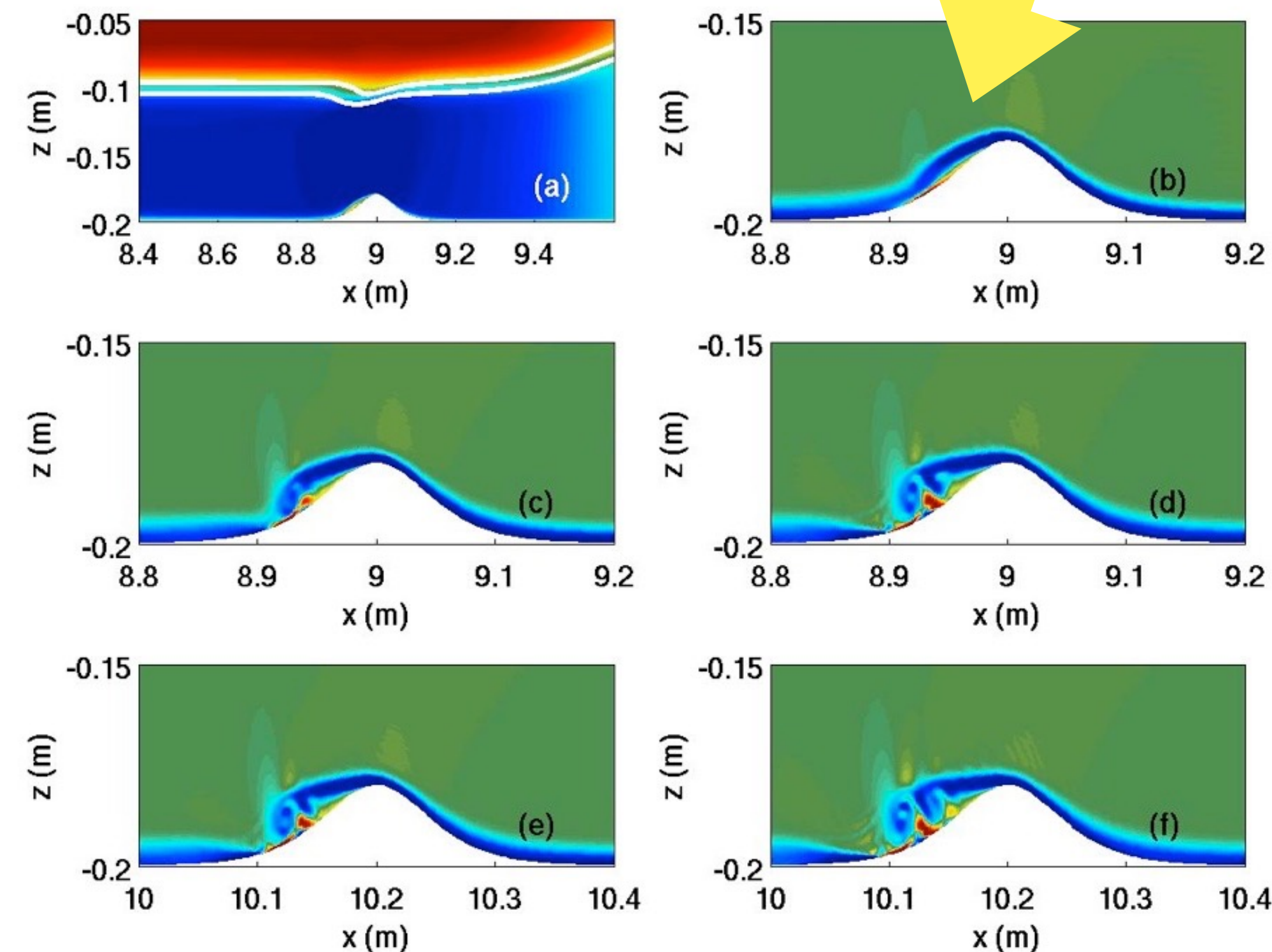


- ❖ Boundary Layers don't always stay confined to the solid boundary.
- ❖ This process is called separation and has profound effects on aerodynamics (stall), sports (curling a free kick around a wall) and other applied fields.



❖ Separation is very difficult to address theoretically (triple deck theory).

❖ Flow separation leads to vorticity production (*top left panel shows coloured horizontal velocity*) other panels show vorticity) and breakdown of flow into turbulence



A “good” simulation of a boundary layer will be both higher order than just first or second order and will have extra resolution in the boundary layer to ensure features are resolved.